Occupational sitting and health risks: a systematic review.

CONTEXT: Emerging evidence suggests that sedentary behavior (i.e., time spent sitting) may be negatively associated with health. The aim of this study was to systematically review the evidence on associations between occupational sitting and health risks. EVIDENCE ACQUISITION: Studies were identified in March-April 2009 by literature searches in PubMed, PsycINFO, CENTRAL, CINAHL, EMBASE, and PEDro, with subsequent related-article searches in PubMed and citation searches in Web of Science. Identified studies were categorized by health outcome. Two independent reviewers assessed methodologic quality using a 15-item quality rating list (score range 0-15 points, higher score indicating better quality). Data on study design, study population, measures of occupational sitting, health risks, analyses, and results were extracted. EVIDENCE SYNTHESIS: 43 papers met the inclusion criteria (21% cross-sectional, 14% case-control, 65% prospective); they examined the associations between occupational sitting and BMI (n=12); cancer (n=17); cardiovascular disease (CVD, n=8); diabetes mellitus (DM, n=4); and mortality (n=6). The median study-quality score was 12 points. Half the cross-sectional studies showed a positive association between occupational sitting and BMI, but prospective studies failed to confirm a causal relationship. There was some case-control evidence for a positive association between occupational sitting and cancer; however, this was generally not supported by prospective studies. The majority of prospective studies found that occupational sitting was associated with a higher risk of DM and mortality. CONCLUSIONS: Limited evidence was found to support a positive relationship between occupational sitting and health risks. The heterogeneity of study designs, measures, and findings makes it difficult to draw definitive conclusions at this time.